

Arya Pratap Singh

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EDUCATION

Indian Institute of Information Technology Ranchi

B.Tech. in Computer Science and Engineering, CGPA: 8.06/10

Ranchi, Jharkhand

Aug 2023 -- May 2027

DAV International School

Senior Secondary (CBSE), Science, Class XII: 95.6%, Class X: 98.2%

Mumbai, Maharashtra

2021 -- 2023

EXPERIENCE

Software Engineer Intern

Global Groupware Solutions Ltd.

Jun 2026 -- Present

Remote

- Owned major parts of the MyEmployWise 1.0 to 2.0 migration across frontend architecture, build infrastructure, platform services, and production engineering in a multi-application **Nx monorepo**.
- Migrated **8 production applications** from Webpack and Babel to **Rspack and SWC**, reducing full production build time from 349.70s to 158.83s and making builds **54.6% faster** through transpilation, dependency, and build-pipeline optimizations.
- Implemented route-level code splitting, React.lazy, Suspense, dynamic imports, shared-library optimizations, and dependency pruning, reducing initial JavaScript by **24.4% raw and 29.8% gzip**.
- Contributed to the architecture of the backend **Policy Evaluation Manager (PEM)**, a metadata-driven platform service that evaluates tenant-configured policies and transactional rules at runtime using Transaction Feature Templates (TFTs) and Entity Templates (ETTs).

Software Engineer Intern

StealthEra Innovations Pvt. Ltd.

May 2026 -- Aug 2026

Remote

- Architected a real-time wearable telemetry platform spanning iWOWN smartwatch hardware, backend ingestion services, **Redis**, React analytics dashboards, and a Flutter application while owning major system architecture and engineering decisions.
- Built deployment services and **Docker** images for all three production services, operated the initial platform on **AWS**, and later led its migration to **GCP** with a redesigned data architecture using big data and time-series storage for real-time telemetry workloads.
- Optimized heart-rate and blood-oxygen (SpO2) processing and synchronization, reducing wearable-to-dashboard propagation from approximately **16 minutes to under 1.37 minutes**, a latency reduction of more than **91%**.
- Implemented backend **rate limiting and request safeguards** across service endpoints to control burst traffic, protect backend resources, and improve service stability under load.

Full Stack Developer Intern

Acumensa Technologies Pvt. Ltd.

Nov 2024 -- Apr 2025

Remote

- Built distributed browser-automation and web-scraping infrastructure using **Playwright, Selenium, Docker, Kubernetes, Amazon EKS, and DigitalOcean Kubernetes (DOKS)**, supporting pipelines processing more than **10M records**.
- Provisioned and managed infrastructure with **Terraform**, creating repeatable Kubernetes deployments, Docker images, deployment scripts, worker automation, and horizontal scaling workflows across multiple scraper systems.
- Used **Prometheus, Grafana, and LGTM observability tooling** to monitor scraper throughput, execution latency, success and failure rates, worker utilization, and container resource usage; built automated validation and failure-detection tooling for large-scale scraping workloads.

PROJECTS

EduBox: Intelligent Student Workspace | Next.js, TypeScript, Convex, Nuclia

Aug 2025 -- Sep 2025

- Built a real-time full-stack workspace with document ingestion, synchronization, analytics, a serverless Convex backend, and a **Retrieval-Augmented Generation (RAG)** pipeline using Nuclia, Gemini, Groq, and CopilotKit.

AI Video Generation Platform | Next.js, PostgreSQL, Remotion

Sep 2024 -- Nov 2024

- Built an open-source multi-stage media-processing pipeline integrating LLM inference, speech synthesis, transcription, persistence, and programmatic rendering, earning **110+ GitHub stars and 29 forks**.

Mindful: AI Mental Wellness App | Flutter, Node.js, PostgreSQL, Redis

Oct 2024 -- Nov 2024

- Designed a Flutter and Node.js application using PostgreSQL, Redis, multiple AI providers, Clean Architecture, and a TensorFlow emotion-recognition model, with backend services deployed on Render.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS, Amazon EKS, GCP, DigitalOcean, Kubernetes, Docker, Terraform, Infrastructure as Code (IaC)

DevOps & Observability: CI/CD Pipelines, GitHub Actions, Prometheus, Grafana, LGTM, Monitoring, Linux, Git, Rate Limiting

Platform & Build Engineering: Nx, Rspack, SWC, Webpack, Build Automation, Bundle Analysis

Systems & Data: Apache Kafka, Redis, PostgreSQL, REST APIs, Playwright, Selenium, Web Scraping

Languages & Frameworks: Python, Go, TypeScript, JavaScript, Bash, SQL, C/C++, Express.js, Hono.js, FastAPI, React, Next.js